

## 3.1 Rational Numbers and the Number Line – Part 1

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                        |                                                                                   |                                                                                                                       |                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context.                                                                                                                                   |                                                                                   |  Learning Targets                  | <ul style="list-style-type: none"><li>I can demonstrate an understanding of negative numbers.</li><li>I can recognize that a number in the real-number system can have a variety of classifications.</li><li>I can plot rational numbers on a horizontal and vertical number line.</li></ul> |
|  Benchmark Coherence | Building On<br>N/A                                                                                                                                                                                                                                                                                                                                     |                                                                                   | Working Toward<br>N/A                                                                                                 |                                                                                                                                                                                                                                                                                              |
|  Essential Questions | <ul style="list-style-type: none"><li>How could you use a number line to represent positive and negative values relative to 0?</li><li>How does the direction, or how far left or how far right of 0, relate to the value of a number on a number line?</li><li>How can you scale number lines with tick marks to include fractional values?</li></ul> |                                                                                   |                                                                                                                       |                                                                                                                                                                                                                                                                                              |
|  Lesson Narrative    | Students develop an understanding of negative numbers and their placement on the number line. Students explore the components of the real-number system. Students then plot rational numbers on the number line.                                                                                                                                       |                                                                                   |                                                                                                                       |                                                                                                                                                                                                                                                                                              |
|  Component Summary | 3.1.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                | Create an equation equal to zero using positive whole numbers ( <i>SE p. 75</i> ) | 3.1.5 Bring It Together (10 min.)                                                                                     | Guided practice plotting rational numbers on a number line ( <i>SE p. 79</i> )                                                                                                                                                                                                               |
|                                                                                                       | 3.1.2 Exploration (5 min.)                                                                                                                                                                                                                                                                                                                             | Explore the existence of negative numbers ( <i>SE p. 76</i> )                     | 3.1.6 Try It (5-10 min.)                                                                                              | Independent practice plotting rational numbers on horizontal and vertical number lines ( <i>SE p. 80</i> )                                                                                                                                                                                   |
|                                                                                                       | 3.1.3 Guided Instruction (5-10 min.)                                                                                                                                                                                                                                                                                                                   | Learn about the real-number system ( <i>SE p. 76</i> )                            | 3.1.7 Your Turn! (5 min.)                                                                                             | Additional practice plotting rational numbers on horizontal and vertical number lines ( <i>SE p. 81</i> )                                                                                                                                                                                    |
|                                                                                                       | 3.1.4 Activity (5-10 min.)                                                                                                                                                                                                                                                                                                                             | Extend a horizontal number line to include negative values ( <i>SE p. 78</i> )    | 3.1.8 Wrap-Up (~5 min.)                                                                                               | Demonstrate understanding of negative numbers ( <i>SE p. 81</i> )                                                                                                                                                                                                                            |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                                               |                                                                                   | Materials                                                                                                             |                                                                                                                                                                                                                                                                                              |
|                                                                                                       | <u>Key Terms:</u> <ul style="list-style-type: none"><li>Natural numbers</li><li>Whole numbers</li><li>Integers</li><li>Rational numbers</li><li>Real numbers</li></ul> <u>Additional Terms:</u> N/A                                                                                                                                                    |                                                                                   | <ul style="list-style-type: none"><li>(Optional) Different types of paper (patty, lined, graph, chart)</li></ul>      |                                                                                                                                                                                                                                                                                              |
|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                        |                                                                                   | Additional Preparation                                                                                                |                                                                                                                                                                                                                                                                                              |
|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                        |                                                                                   | <ul style="list-style-type: none"><li>3.1.2 Exploration contains a video to accompany the question prompts.</li></ul> |                                                                                                                                                                                                                                                                                              |

|                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher Tips               | <b>Common Misconceptions</b> <ul style="list-style-type: none"><li>Students may interpret larger negative numbers as having a larger value.</li><li>Students may believe negative numbers cannot be rational (because they are negative).</li><li>Students may have difficulty making the tick marks for the <math>\frac{1}{4}</math> intervals.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Things to Consider</b> <ul style="list-style-type: none"><li>Explain and demonstrate that the negative numbers are a mirror image of the positive numbers on the number line to help clarify student thinking.</li><li>3.1.5 Bring It Together is intended to be guided after students have had an opportunity to explore the number line in the 3.1.4 Activity. Consider the language used when modeling how to plot values on the number line, particularly when plotting negative values, which is new for students. Use a think aloud to model the thinking involved so students can apply similar reasoning when practicing on their own in the components that follow.</li><li>If time is an issue, consider assigning 3.1.7 Try It! As homework.</li></ul> |
|                                                                                                                 | <b>Literacy and Language Routines</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Mathematical Thinking and Reasoning (MTR) Standard</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                 | <b>Stronger and Clearer Each Time</b><br>3.1.2 Exploration provides an opportunity to practice the Stronger and Clearer Each Time routine. After developing individual responses, students can work in small groups to critique each other’s ideas about negative numbers. Critiques should include suggestions for improving the clarity of each other’s verbal and written explanations.                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>MA.K12.MTR.4.1: Discussion and Reflection</b><br>Students will work to effectively communicate mathematical ideas and vocabulary regarding negative numbers. In the 3.1.3 Guided Instruction, students can share their responses and discuss ideas and thoughts about the real-number system. Students should be encouraged to provide justification for responses.                                                                                                                                                                                                                                                                                                                                                                                               |
| <br>Differentiate Instruction | Math is what connects us all. Consider adding variety to 3.1.4 Activity with a human number line. Students can choose a number special to them between 0 and 20. Students should choose their number and determine whether they feel “negative” or “positive” to determine which side of the number line they will stand. Students are then challenged to make decisions for how “far away” they should stand from each other to represent the mental work of scaling the number line. This kinesthetic learning experience can be a fun and engaging way to begin developing students’ number senses using a number line. Note that incorporating this activity will require additional time. Consider when you would want to incorporate it and how you might adjust instruction to accommodate the additional time. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Lesson Notes:                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

### 3.1.1 Warm-Up: Creating Zero

**Component Overview:** Students use whole numbers one to nine to create an equation equal to zero. Students then switch equations with a partner. Partners check each other's equations and discuss their strategies.



#### 3.1.1 Warm-Up: Creating Zero

1. Use only digits 1 to 9, at most one time each, so that the equation is true.

$$(\square + \square - \square + \square) - (\square + \square - \square + \square) = 0$$

*Answers will vary.*

*Possible solution:*  $(9 + 1 - 8 + 5) - (7 + 4 - 6 + 2) = 0$

2. Switch your answer with a partner to check. Ask your partner their strategy for finding their answer and write their strategy below.

*Answers will vary.*



*Require students to provide reasoning for methods used to solve the equation. Encourage students to clarify and critique during discussions. Students may change responses based on partner discussions.*

## 3.1.2 Exploration: How Can Negative Numbers Exist?

**Component Overview:** Students watch “Negative Numbers: An Overview.” Students explore graphing negative numbers on a number line and real-world examples of negative numbers.



### 3.1.2 Exploration: How Can Negative Numbers Exist?

Negative numbers are numbers less than zero.  
Examples include:

- a temperature below 0 degrees
  - the amount of money owed, or debt
  - a contestant’s score on Jeopardy (when someone misses a question, he/she loses points, so his/her score could be a negative amount)
1. Discuss the examples of negative numbers with your partner.
  2. Discuss with your partner which of the contexts in the table could have a value less than 0, then write an example or explain your thinking.

| Context          | Example or Explanation                                                                                            |
|------------------|-------------------------------------------------------------------------------------------------------------------|
| Direction        | <i>Answers will vary. An example would be if you are driving somewhere, but you drove in the wrong direction.</i> |
| Points in a game | <i>Answers will vary. An example is losing points in a game.</i>                                                  |
| Time             | <i>Answers will vary. I don’t think it’s possible to have time less than 0.</i>                                   |



Playing the two-minute video “Negative Numbers: An Overview” prior to starting the activity provides visual and auditory entry points. Pause the video to explain that Mount Everest is so cold because it is the highest mountain above sea level. Students may also benefit from comparing and contrasting what they know about the weather in Alaska, Idaho, and Florida.



**Stronger and Clearer Each Time**  
Consider requiring students to develop written responses prior to partner and/or group discussions. Then, have students provide each other with feedback. Next, students can revise their responses. Repeat this process as time allows for increasingly precise answers.

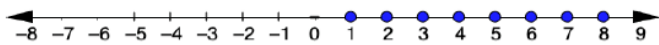
### 3.1.3 Guided Instruction: The Real Number System

**Component Overview:** Students learn about the sets of numbers in the real-number system. A visual entry point is provided as students observe where the natural numbers, whole numbers, and integers lie on the number line. Then, students explore examples of natural numbers, integers, and rational numbers.



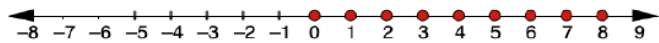
#### 3.1.3 Guided Instruction: The Real Number System

The values in blue on the number line are some of the numbers in the set of **natural numbers**.



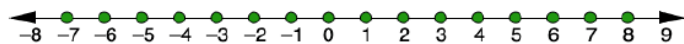
**Natural numbers** are the counting numbers  
{1, 2, 3, 4, 5 ...}.

The values in red on the number line are some of the numbers in the set of **whole numbers**.



**Whole numbers** are natural numbers and zero.  
{0, 1, 2, 3, 4, 5 ...}

The values in green on the number line are some of the values in the set of **integers**.



**Integers** are whole numbers and their opposites.  
{... -4, -3, -2, -1, 0, 1, 2, 3, 4 ...}



What do you notice about the numbers as you move to the left of zero? How does the value of numbers compare as you move to the left of zero versus as you move to the right of zero?

1. Use the number lines to help to answer each of the following questions.
  - a. What do you notice about the set of natural numbers and the set of whole numbers?  
*Answers will vary. Only positive numbers are included in natural numbers. The natural numbers are whole numbers except for 0.*

### 3.1.3 Guided Instruction: The Real Number System (continued)

- b. What do you notice about the set of integers?  
*Answers will vary. Positive and negative numbers are included.*
- c. What type of number is  $-10$ ?  
*Integer*
- d. What type of number is  $27$ ?  
*Natural Number, Whole Number, and Integer.*
- e. Describe the numbers that appear in all three sets.  
*Answers will vary. Natural numbers, or positive whole numbers, appear in all three sets.*

Camaria wondered to herself, "What about numbers such as  $3\frac{1}{2}$  and  $-1.75$ ?"

These numbers are examples of **rational numbers**.



**Rational numbers** are numbers that can be expressed as the ratio of two integers.

2. Use the definition of rational numbers to write each as a ratio of two integers.

| Num<br>ber | Ratio of<br>Two<br>Integers |
|------------|-----------------------------|
| $-1.5$     | $-\frac{3}{2}$              |
| $9$        | $\frac{9}{1}$               |

In your study of mathematics, you will learn about **irrational numbers**, which are numbers that cannot be written as a ratio of two integers.



The set of **real numbers** is the set of all rational and irrational numbers.



Ask students to share the ratios they wrote. Look for responses that have typical misconceptions. Some examples could be  $\frac{-1.5}{1}$  or  $\frac{9}{9}$ . Also, consider sharing ratios that students may not have written such as  $\frac{9}{-4}$  or  $\frac{18}{2}$ .

### 3.1.4 Activity: Horizontal Number Line

**Component Overview:** Students plot rational numbers on a number line. This activity provides a visual entry point as students develop a conceptual understanding of negative numbers.



#### 3.1.4 Activity: Horizontal Number Line

Using a sheet of paper, follow the steps to make your own number line.

- Step 1: Use a straightedge or a ruler to draw a horizontal line. Mark the middle point of the line and label it 0.
- Step 2: To the right of 0, draw tick marks that are 1 centimeter apart. Label the tick marks 1, 2, 3...10.
- Step 3: Fold your paper so that a vertical crease goes through 0 and the two sides of the number line match up perfectly.
- Step 4: Use the fold to help you trace the tick marks that you already drew onto the opposite side of the number line. Unfold and label the tick marks  $-1, -2, -3, \dots, -10$ . This represents the negative side of your number line.

Use the number line to answer each of the following questions.

- Describe how far away the number 5 is from 0.

*The number 5 is **five** units to the **right** of 0 on the number line.*

- Write a positive number and a negative number where each is farther away from 0 than the number 5.

*Answers will vary.*

*Example:  $-6$  and  $6$*

- Describe how far away the number  $-2$  is from 0.

*The number  $-2$  is **two** units to the **left** of 0 on the number line.*

- Write a positive number and a negative number that are each farther away from 0 than  $-2$ .

*Answers will vary.*

*Example:  $-3$  and  $3$*



Visit the “Differentiate Instruction” note in Additional Planning Support for the Human Number Line activity. If using this activity, consider how you will adjust instruction to accommodate the additional time it would require.



Different types of paper can be used to answer practice problems. Lined, chart, and graph paper simplify making a number line, because they already include tick marks. Patty paper simplifies making a number line, because the positive side of the number line can be folded and rubbed onto the negative side of the number line.

### 3.1.5 Bring It Together: Plotting Rational Numbers on Number Lines

**Component Overview:** Students learn to plot positive and negative numbers on a number line. Then, students extend their knowledge and skills to include plotting fractions and decimals on the number line. Teacher should guide students through this component, modeling how to think through the relationships between the values to determine each value's location.



#### 3.1.5 Bring It Together: Plotting Rational Numbers on Number Lines

When plotting numbers on a number line, the sign of the number indicates which direction from 0 to move on the number line.

- On a horizontal number line, positive numbers are

to the 
☐ right  
☐ left
  of zero.

- On a vertical number line, positive numbers are

☐ above  
☐ below
  zero.

- On a horizontal number line, negative numbers are

to the 
☐ right  
☐ left
  of zero.

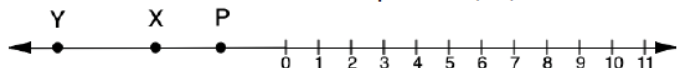
- On a vertical number line, negative numbers are

☐ above  
☐ below
  zero.



Students can use the folded number line they created during the previous component as an additional scaffold when trying to understand how to locate each value on the number line, particularly when plotting negative values.

- A number line is shown with points  $P$ ,  $X$ , and  $Y$ .



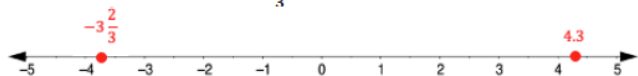
- What type of numbers are  $P$ ,  $X$ , and  $Y$ ?  
*Negative numbers*
- Estimate the location of points  $P$ ,  $X$ , and  $Y$ .  
*-2, -4, -7*



### 3.1.5 Bring It Together: Plotting Rational Numbers on Number Lines (continued)

2.

- a. Plot the values  $-3\frac{2}{3}$  and 4.3 on the number line.



- b. How did the tick marks between each integer help you determine where to plot 4.3?

*Answers will vary. The tick marks show integers and the halfway point between them. I know 4.3 is between 4 and 4.5, but closer to 4.5, so that was helpful to plot 4.3.*

- c. How did the tick marks between each integer help you determine where to plot  $-3\frac{2}{3}$ ?

*Answers will vary. I know  $-3\frac{2}{3}$  is between  $-3$  and  $-4$  on the number line, but closer to  $-4$ .  $-3\frac{2}{3}$  is also farther to left of 0 than  $-3.5$ , so I used that to help me estimate its location.*



*How can making a tick mark that is halfway between each value on the number line help you plot fractions and decimals? How can you convert between fractions and decimals to assist in comparing their values?*



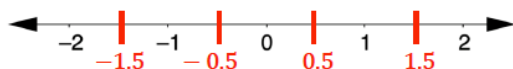
**Think Aloud**

*For question 2, the teacher should think aloud as they plot the values on the number line to model the type of thinking students can use when making these decisions on their own in the next component.*

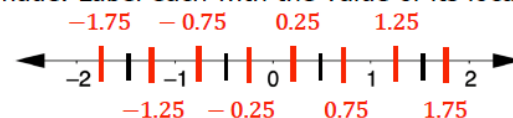
*Based on student understanding to this point, the teacher may decide to release question 3 to students or continue modeling to ensure better understanding of the new content before releasing in the next component.*

3.

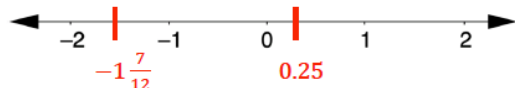
- a. On the number line, make a tick mark that is halfway between each value on the number line. Label each tick mark with the value of its location.



- b. Make another tick mark halfway between each value and the tick marks that were previously made. Label each with the value of its location.



- c. Plot the values  $-1\frac{7}{12}$  and 0.25 on the number line.



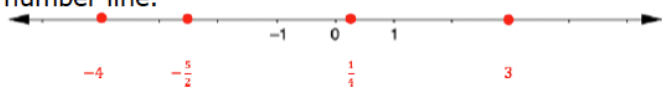
## 3.1.6 Try It: Plotting Rational Numbers on a Number Line

**Component Overview:** Students, independently or with a partner, practice plotting integers, fractions, and decimals on horizontal and vertical number lines. Circulate while students work to support struggling students, monitor progress, and select which students you wish to have share during the debrief.

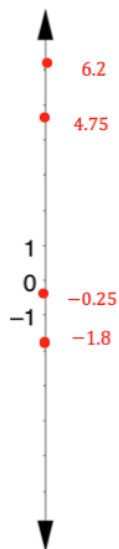


### 3.1.6 Try it: Plotting Rational Numbers on a Number Line

- Plot the numbers  $-\frac{5}{2}$ ,  $-4$ ,  $\frac{1}{4}$ , and 3 on the horizontal number line.



- Plot the rational numbers  $-0.25$ ,  $-1.8$ ,  $6.2$ , and  $4.75$  on the vertical number line.



What type of numbers are below/to the left of zero? What type of numbers are above/to the right of zero?

## 3.1.7 Your Turn!

**Component Overview:** Students independently practice plotting integers, fractions, and decimals on horizontal and vertical number lines. Students are provided with a visual entry point through the use of number lines. If time is an issue, then consider assigning 3.1.7 Your Turn! for homework.



### 3.1.7 Your Turn!

- Negative numbers are ☐ greater ☒ less than zero. On a horizontal number line, negative numbers are to the ☐ right ☒ left of zero. On a vertical number line, negative numbers are ☐ above ☒ below zero.

- Plot the integers 2, -3, 0, -5 and 5 on the horizontal number line.



- Plot the rational numbers  $-1\frac{1}{5}$ ,  $-\frac{4}{5}$ , 2, and  $\frac{6}{5}$  on the vertical number line.



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

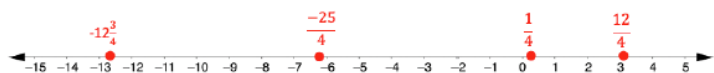
## 3.1.8 Wrap-Up: Ticket Out the Door

**Component Overview:** In this formative assessment, students demonstrate their understanding of plotting rational numbers on a number line. Use information from 3.1.8 Wrap-Up to decide which students need remediation before the next lesson.

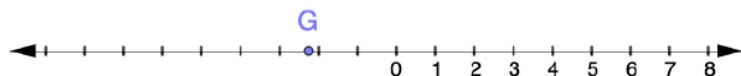


### 3.1.8 Wrap-Up: Ticket Out the Door

- Plot the rational numbers  $-\frac{25}{4}$ ,  $-12\frac{3}{4}$ ,  $\frac{12}{4}$ , and  $\frac{1}{4}$  on the number line.



- Explain how to estimate the value of  $G$ .



Answers will vary.  $G$  is to the left of zero, so it will be a negative value. It's between  $-2$  and  $-3$ , but closer to  $-2$ , so I would estimate its value is about  $-2.25$ .



How can converting fractions to decimals or vice versa help you determine where to place rational numbers on a number line?